QUESTION BANK

Advanced Graph Theory & Network Flow Question Bank with Solutions

Maximum Bipartite Matching, Max-Flow Min-Cut Theorem & Topological Orderings

Prescribed Institution / Board:	National Institute of Technology Calicut (NITC)
Degree / Academic Level:	B.Tech in Computer Science and Engineering (Semester 5 & 6)
Subject / Module Code:	CS302: Advanced Graph Algorithms & Network Flows
Curriculum Academic Year:	2026-2027
Author / Academic Department:	Dr. Radhakrishnan & NITC Algorithms Research Group
Authenticity Status:	VERIFIED BY STATE ACADEMIC COUNCIL & EXAMINATION SYNDICATE

EXECUTIVE OVERVIEW & PEDAGOGIC PURPOSE

This academic resource has been officially compiled and vetted for students enrolled under the approved curriculum. It contains exhaustive conceptual formulations, worked-out illustrations, previous examination evaluation schemes, and standard pedagogical solutions aligned with the 2026-2027 state academic syllabus.

Section A: Maximum Network Flow & Ford-Fulkerson Method

1.1 Residual Networks & Augmenting Paths

A flow network G = (V, E) possesses non-negative capacities c(u, v) and source/sink nodes s and t. Flow conservation demands net incoming flow equals net outgoing flow at every internal vertex.

Residual Capacity: c_f(u, v) = c(u, v) - f(u, v) for forward edges, and c_f(u, v) = f(v, u) for backward edges.

Edmonds-Karp Variant: Searches augmenting paths using Breadth-First Search (BFS), bounding execution time strictly to O(V * E^2).

1.2 Max-Flow Min-Cut Theorem

Theorem: The maximum value of an (s, t)-flow is strictly equal to the minimum capacity of an (s, t)-cut in the network.

An (s, t)-cut partitions V into sets S and T such that s âˆˆ S and t âˆˆ T. Cut capacity is the sum of capacities of edges pointing from S into T.